September 8, 2022

CURRICULUM VITAE

Leroy L. Jia
Center for Computational Biology
Flatiron Institute, Simons Foundation
162 Fifth Avenue, New York, NY 10010
`royjia117@gmail.com`
`https://royjia117.github.io/`
Phone number available upon request

Citizenship: USA

Positions Held

Flatiron Research Fellow, Center for Computational Biology, Flatiron Institute — 2018-2022
  Supvervisor: Prof. Michael Shelley

Visiting Positions

Visitor, Courant Institute of Mathematical Sciences, New York University — 2019-2020
Visiting Scientist, Department of Physics, Brown University — 2018-2019
Postdoctoral Research Associate, School of Engineering, Brown University — 2018
  Supervisor: Prof. Thomas Powers (1 mo. courtesy appointment)

Education

Ph.D. Applied Mathematics, Brown University — 2018
  Advisors: Profs. Thomas Powers & Robert Pelcovits
  Thesis: "Geometry and Mechanics of Self-Assembling Colloidal Membranes"
Sc.M. Engineering, Brown University — 2017
Sc.M. Applied Mathematics, Brown University — 2013
B.S. Physics, Georgia Institute of Technology — 2012
B.S. Applied Mathematics, Georgia Institute of Technology — 2012
Certificate in I/O Psychology, Georgia Institute of Technology — 2012

Awards and Honors

Sigma Xi, Brown University Chapter — 2018
Outstanding Teaching Assistant — 2014
School of Mathematics MCTP Scholarship, Georgia Institute of Technology — 2012, 2011
Hitohiro Fukuyo Memorial Award, Georgia Institute of Technology — 2011

Research Interests

Applied mathematics: analysis of ODE & PDE, geometry of surfaces, asymptotics, numerics
Theory of soft matter: membranes, fluid interfaces, colloids, liquid crystals, active matter
Biomechanical engineering: complex fluids, elasticity, self-assembly, bio-inspired materials

Publications (* denotes shared first author contribution.)

8. **L. L. Jia**, W. T. M. Irvine, M. J. Shelley. "A boundary integral method for active chiral monolayers." In preparation.
7. J. Robaszewski*, **L. L. Jia***, R. Adkins, R. A. Pelcovits, T. R. Powers, Z. Dogic. "Inducing vesicle formation in colloidal membranes by controlling membrane thickness." To be submitted Fall 2022, preprint available upon request.
6. **L. L. Jia**, W. T. M. Irvine, M. J. Shelley. "Free-boundary problems for odd fluids. I. Formulation and axisymmetric flows." Under review at J. Fluid Mech., preprint available at `arXiv:2202.13962` or upon request.
5. **L. L. Jia**, M. J. Shelley. "The role of monolayer viscosity in Langmuir film hole closure dynamics." J. Fluid Mech. **948** (2022) A1.
4. A. Khanra*, **L. L. Jia***, N. P. Mitchell, A. Balchunas, R. A. Pelcovits, T. R. Powers, Z. Dogic, P. Sharma. "Controlling the shape and topology of two-component colloidal membranes." P. Natl. Acad. Sci. USA. **119**(32) (2022) e2204453119.
3. **L. L. Jia**, S. Pei, R. A. Pelcovits, T. R. Powers. "Axisymmetric membranes under external force: buckling, minimal surfaces, and tethers." Soft Matter **17** (2021) 7268. *Selected as Front Cover.*
2. A. Balchunas*, **L. L. Jia***, M. J. Zakhary, J. Robaszewski, T. Gibaud, Z. Dogic, R. A. Pelcovits, T. R. Powers. "Force-induced formation of twisted chiral ribbons." Phys. Rev. Lett. **125** (2020) 018002.
1. **L. L. Jia**, M. J. Zakhary, Z. Dogic, R. A. Pelcovits, T. R. Powers. "Chiral edge fluctuations of colloidal membranes." Phys. Rev. E **95** (2017) 060701(R).

Teaching Experience

Instructor (Brown University)

- APMA 0160 Introduction to Scientific Computing (Summer 2017). First-year course on computer programming and algorithms. Taught in MATLAB. Rootfinding, numerical linear algebra, finite differences, quadratures, numerical differential equations. I reworked the curriculum to include lectures on optimization and machine learning.
- New Scientist Program Catalyst Content Instructor and Peer Mentor (Summer 2014). NSP Catalyst is a summer program that mentors aspiring STEM majors from traditionally underrepresented groups. My course had online and lecture components and was about complex numbers, differential equations, and linear stability theory.

Teaching Consultant (Brown University)

- Brown University Sheridan Center Teaching Consultant (Fall 2017-Spring 2018). Teaching consultants facilitate workshops and meet with and provide feedback to new TAs.

Graduate Teaching Assistant (Brown University)

- Sheridan Teaching Seminar (Fall 2017). Fundamental teaching and communication strategies. Critical reflection, rhetorical practice, learning design, and active learning.

- APMA 0330 Methods of Applied Math I (Spring 2014). Introduction to differential equations. First and second order ordinary differential equations, complex numbers, inhomogeneous equations, Laplace transforms, series solutions, numerical methods.
- APMA 0340 Methods of Applied Math II (Fall 2014, Spring 2013). Continuation of APMA 0330. Systems of ordinary differential equations, phase space, linear stability, periodic equations, introduction to eigenvalue problems and linear partial differential equations. Recognized as an Outstanding TA in Fall 2014.
- APMA 0350 Honors Methods of Applied Math I (Spring 2015). Rigorous, proof-based version of APMA 0330 intended for mathematics majors.
- Grader for APMA 1360 Topics in Chaotic Dynamics (Spring 2014) and APMA 1650 Statistical Inference I (Fall 2012)

Undergraduate Teaching Assistant (Georgia Institute of Technology)

- MATH 1502 Calculus II with Linear Algebra (Spring 2012, Fall 2011). Second semester calculus (25%) plus elementary linear algebra (75%). Infinite series, Taylor series, vector spaces, matrices, linear transformations, eigenvalues and eigenvectors.
- PHYS 2211 Introductory Physics I (Spring 2012 x3, Spring 2011 x4, Spring 2010 x4, Spring 2009 x1) [multiplier indicates number of sections served that semester]. Calculus-based introductory mechanics course. Kinematics, Newton's laws, momentum, energy, rotational dynamics. Includes a computational lab component.
- PHYS 2212 Introductory Physics (Fall 2011 x2, Fall 2010 x4, Fall 2009 x2) [multiplier indicates number of sections served that semester]. Calculus-based introductory electromagnetism course. Electric and magnetic fields, polarization, circuits, Maxwell's equations, electromagnetic radiation. Includes a computational lab component.
- PHYS 2232 Honors Introductory Physics II (Fall 2011 x1) [multiplier indicates number of sections served that semester]. Faster-paced, more in-depth version of PHYS 2212 for physics majors. Includes a computational lab component.
- Grader for MATH 4107 Abstract Algebra I (Fall 2011, Spring 2011), MATH 4318 Analysis II (Spring 2012), and MATH 4347 Partial Differential Equations I (Fall 2010)

Teaching Workshops Attended

- Course Design Seminar, Sheridan Center for Teaching and Learning (Spring 2018). Explore integrated course design principles and develop inclusive syllabi, assignments, and activities. Covers principles of backwards design, assignment scaffolding, task- and project-based learning, etc. Includes opportunities to create two syllabi.
- Teaching Consultant Program, Sheridan Center for Teaching and Learning (Fall 2017-Spring 2018). Develop and refine skills in peer observation and feedback, leadership, and discussion facilitation. Develop and articulate a teaching philosophy and create a teaching portfolio. Includes eight teaching consultations with new graduate TAs who are observed and given feedback.

- Reflective Teaching Seminar, Sheridan Center for Teaching and Learning (Fall 2016). Develop and refine fundamental teaching and assessment strategies and communication skills based on how students learn. Covers principles of inclusive teaching, backward design, and classroom communication. Includes a teaching observation with feedback.
- Graduate TA Preparation Workshops, Brown University (Fall 2013). Covers rules and guidelines for being an effective graduate teaching assistant.
- Undergraduate TA Preparation Workshops, Georgia Institute of Technology (Fall 2011). Covers rules and guidelines for being an effective mathematics teaching assistant.

Other Teaching Activities

- UNC Chapel Hill Applied Math Graduate Qualifying Exam Tutor (Fall 2021)
- Brown University Math Resource Center Tutor (Fall 2012-Spring 2018)
- Georgia Institute of Technology Math Lab Tutor (Fall 2011-Spring 2012)

Students Mentored

- Research mentor for Steven Pei, undergraduate research (UTRA) student in physics (Fall 2016-Fall 2018). Co-advised with T. Powers. Steven is currently a software engineer at Google.
- Senior project and college mentor for Xingjian "Jackson" Gao, high school student (Summer 2017-Fall 2017). Jackson went on to study CS/applied math at UC Berkeley.
- College mentor for Chia-Yuan "Calvin" Chang, high school student (Summer 2017). Calvin went on to study CS/applied math at UC Davis.

Presentations and Posters

- Seminar, Flatiron Institute Center for Computational Biology Brown Bag Seminar (October 2022)
- Seminar, Manhattan College (August 2022)
- Seminar, High Point University (July 2022)
- Invited talk, Soft Living, Active, and Adaptive Matter Seminar (June 2022)
- Poster, Mechanics of Life Workshop (May 2022)
- Seminar, The Cooper Union (April 2022)
- Seminar, Northwestern University (February 2022)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2021)
- Seminar, Flatiron Institute Center for Computational Biology Brown Bag Seminar (September 2021)
- Contributed talk, APS March Meeting (March 2021)
- Seminar, Flatiron Center for Computational Biology Virtual Retreat (January 2021)
- Seminar, Flatiron Center for Computational Biology Brown Bag Seminar (May 2020)

- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2019)
- Contributed talk, APS March Meeting (March 2019)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2018)
- Poster, Brandeis University, MRSEC Site Visit (May 2018)
- Poster (w/ J. Robaszewski), Brandeis University, MRSEC Site Visit (May 2018)
- Poster (w/ A. Balchunas), Brandeis University, MRSEC Site Visit (May 2018)
- Invited talk, SIAM Conference on the Life Sciences (August 2018)
- Seminar, Flatiron Institute (May 2018)
- Seminar, Brown University (May 2018)
- Seminar, UMass Amherst (April 2018)
- Contributed talk, APS March Meeting (March 2018)
- Contributed talk, APS March Meeting (March 2017)
- Poster, Brown University, Department of Physics Poster Session (November 2016)
- Seminar, Brandeis University (October 2016)
- Contributed talk, New England Complex Fluids Workshop (September 2016)
- Contributed talk, APS March Meeting (March 2016)
- Seminar, Brown University Applied Math Graduate Seminar (March 2016)
- Presentation, Brown University Applied Math Graduate Retreat (September 2015)
- Seminar, Brandeis University (August 2015)
- Poster, UC Boulder, Boulder Summer School in Condensed Matter Physics (July 2015)
- Presentation, Kobe University (August 2014)

Schools and Workshops Attended

- Mechanics of Life Workshop, Flatiron Institute (May 2022)
- Flatiron-wide Algorithms and Mathematics, Flatiron Institute (October 2021)
- The Physics of Elastic Films: from Biological Membranes to Extreme Mechanics, Kavli Institute for Theoretical Physics (May-June 2021)
- Flatiron-wide Algorithms and Mathematics, Flatiron Institute (October 2020)
- Universality: Turbulence Across Vast Scales, Flatiron Institute (December 2019)
- Flatiron-wide Algorithms and Mathematics, Flatiron Institute (October 2019)
- Supercomputing in Plain English, University of Oklahoma Supercomputing Center for Education and Research (Spring 2018)
- Boulder Summer School in Condensed Matter Physics, UC Boulder (July 2015)
- Summer School on Soft Solids and Complex Fluids, UMass Amherst (June 2015)

- Brown-ICERM-Kobe Simulation Summer School, Brown University/RIKEN/Kobe University, (August-September 2014)

Diversity, Outreach, and Service Activities

- Contract Editor, American Journal Experts (August 2022 - Present)
- Simons Foundation Asian American & Pacific Islander Employee Resource Group (Winter 2021-present)
- Flatiron Institute Professional Development Program Committee (Winter 2021-present)
- KITP Films '21 Advice for PhD Students Panelist (June 2021)
- Brown University New Teaching Assistant Orientation Facilitator (Fall 2017)
- Brown University Applied Math Grad-Undergrad Mentor (Spring 2016-Spring 2018)
- Brown University New International Graduate Student Orientation Peer Mentor (Fall 2016, Fall 2017)
- Brown Applied Math Graduate Retreat Project Leader (Fall 2016). My project was about the Euler-Plateau problem.
- Mandarin-English Language Exchange Volunteer Teacher (Spring 2013-Spring 2018) and Japanese-English Language Exchange Volunteer Teacher (Spring 2018-Fall 2020)
- Team Leader for GT 1000 Freshman Seminar, engineering section (Spring 2011)
- Team Leader for GT 1000 Freshman Seminar, physics section (Fall 2010)
- Volunteer math tutor for ESOL middle school students (Fall 2007-Spring 2008)

Personal

- Citizenship: USA (*jus soli*)
- Technical: MATLAB, Mathematica, C++, Python, Java, R, HTML, HPC, Git, 3D printing training from Brown Design Workshop, Blender
- Languages: Mandarin Chinese (fluent), Japanese (conversational), Spanish (basic)
- Memberships: APS (DFD & DSOFT), SIAM

References

Mr. Joseph Browne [teaching reference]
Site Director, Northeastern at SMASH
joseph.browne@smash.org
(617)-229-5639

Prof. Robert Pelcovits
Department of Physics, Brown University
robert_pelcovits@brown.edu
(401)-863-1432

Prof. Thomas Powers
School of Engineering & Department of Physics, Brown University
thomas_powers@brown.edu
(401)-863-2868

Prof. Michael Shelley [possibly slow to respond to letter requests]
Center for Computational Biology, Flatiron Institute
Courant Institute of Mathematical Sciences, New York University
mshelley@simonsfoundation.org
(212)-998-3284